Drinking Water or Municipal Water

Water is a vital body fluid and is essential for all living creatures. Human beings consume a considerable amount of water for drinking and other domestic purposes.

Standard Specification of Drinking Water

Municipal water supply standards are two types

i) **Primary Standards :-**

Those are the standards which specify the (MCL) of various dissolved minerals based on their effect on human health.

ii) **Secondary Standards :-**

Secondary standards may vary from place to place depending upon the taste, odour, colour and hardness of water etc

Requisites of Drinking Water

1. It should be clear, colourless and odourless.
2. It should be good in taste.
3. Its pH should be in the range of 7.0 - 8.5.
4. It should be free from disease producing bacteria.

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Treatment of Water for Municipal Supply

Process used for Removal Impurities

1) Screening – Floating material such as leaves and twigs.

2) Sedimentation – Suspended impurities such as sand, clay

3) Coagulation – Fine suspended inorganic matter

4) Filtration – colloidal impurities and large organism

5) Disinfection – Pathogenic bacteria

Screening

It is a process of removing floating materials like leaves and twigs etc. Raw materials is passed through screens having holes, when the floating matter is retained by them and water is allowed to pass.

Sedimentation

This is the process of removing suspended impurities by allowing the water to stand undisturbed in big tanks for 2-8 hours.

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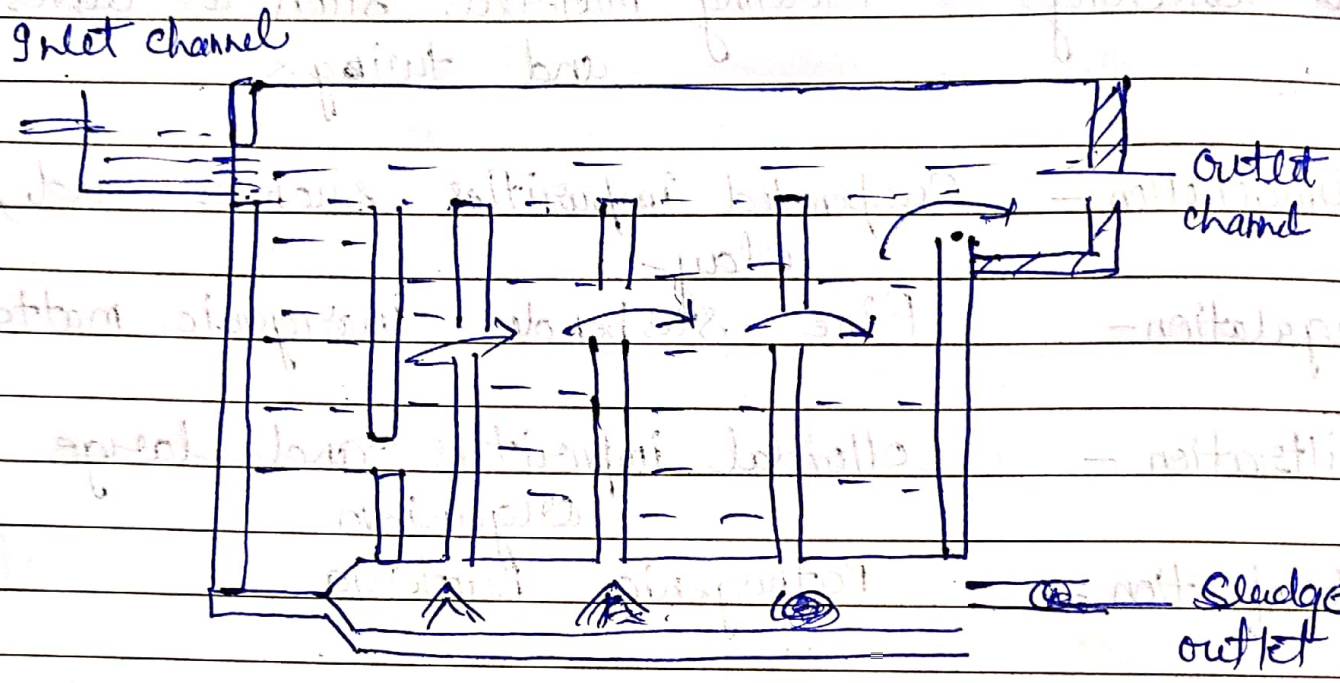
Most of the suspended particles settle down at the bottom of the tank due to the force of gravity. This process removes only 70-75% of the suspended matter.



Sedimentation with Coagulation

When water contains finely divided silica, clay and organic matter do not settle down easily and cannot be removed by sedimentation. Coagulants like alum or ferrous sulphate produce Al^{+3} or Fe^{+2} which neutralize the oppositely charged colloidal and clay particles. After losing their charges, these particles come nearer to one another and combine to form bigger particles which settle down due to the force of gravity.

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(i)

$$Al_2(SO_4)_3 + 3Ca(HCO_3)_2 \longrightarrow 2Al(OH)_3 \downarrow + 3CaSO_4 + 6CO_2 \uparrow$$

(ii) Ferrous Sulphate – [$FeSO_4 \cdot 7H_2O$]

$$FeSO_4 + Mg(HCO_3)_2 \longrightarrow Fe(OH)_2 \downarrow + MgCO_3 + CO_2 + H_2O$$

Filteration

This is the process of removal of impurities (colloidal material, suspended matter etc) and some of micro organism by passing water through a porous material consisting of a bed of fine sand and other granular material.

The filters used for water treatment are of two types i) Gravity filters ii) pressure filters.

Sterilization (disinfection) of water

The process of destroying / killing of pathogenic bacterias and other micro-organism from water to make it safe for use is known as disinfection and the chemicals used for this purpose are called disinfectants.

OR

Sterilization means complete destruction of all living organism which is possible by boiling the

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water over a period of time.

Disinfection of water

i) Boiling of water
ii) Bleaching powder method
iii) Chlorination method
iv) Ozonization process
v) $KMnO_4$ method
vi) By ultra violet light

Bleaching Powder Method

It is commonly represented as $CaOCl_2$. It react with water to produce hypochlorous acid. 1 kg of bleaching water is sufficient to stoilize one million litrus of water.

$$CaOCl_2 + H_2O \longrightarrow Ca(OH)_2 + Cl_2$$

$$Cl_2 + H_2O \longrightarrow HCl + HOCl$$

Hypochlorous acid

Chlorination Method

It can be applied to water in liquid or gaseous form. In these days liq. chlorine is usually preferred. It produces Hypo-chlorous acid, which is a powerful germicide

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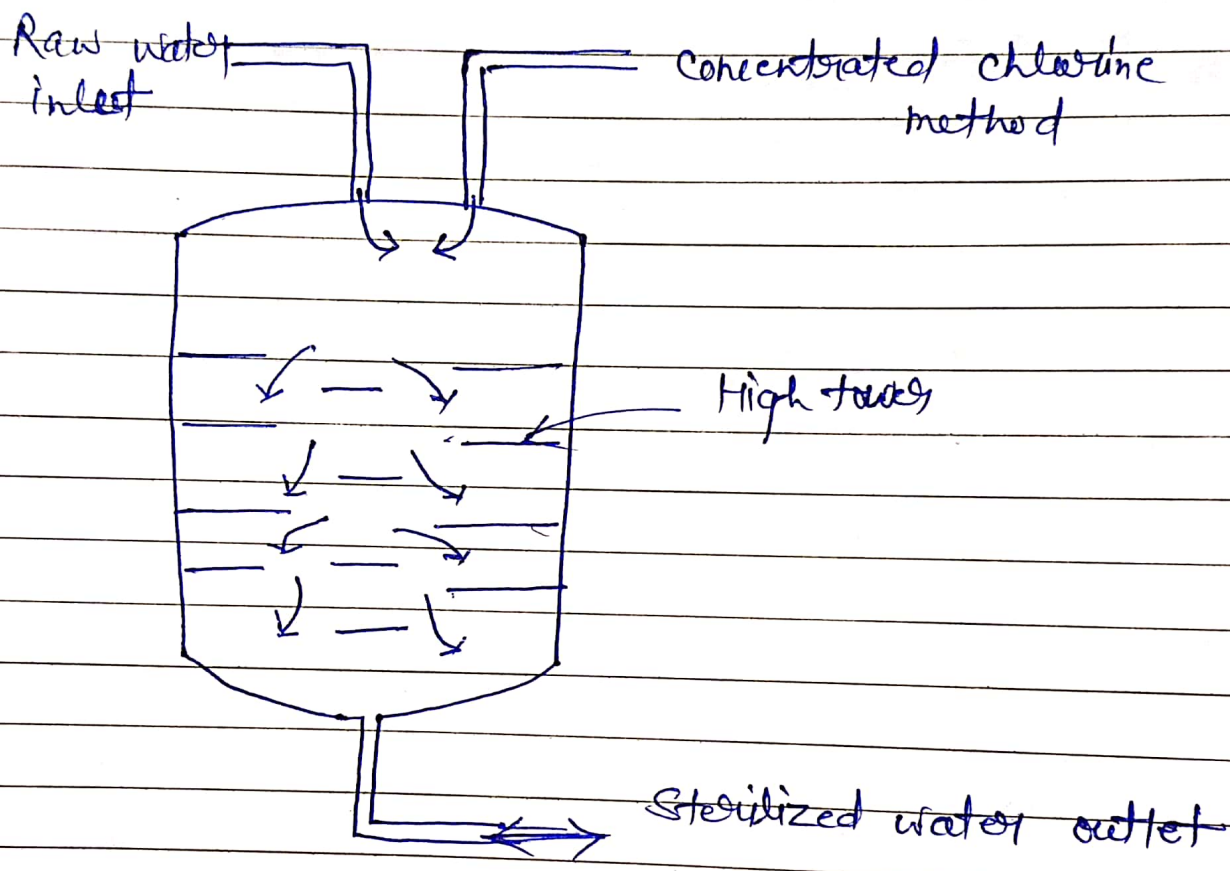
$Cl_2 + H_2O \longrightarrow HOCl + HCl$

$HOCl \longrightarrow$ Bacterias are killed



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